

# Lockout steps

| Lockout phases     | Nature of operations                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    | Electric                                                                                                                                                                                                                                                                                                                                                                                              | Mechanical                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                    |                                                                                                                                                                                                                                                                                                                                                                                                       | Machine stopped / moving                                                                                                                                                                                                                                                                                      | Pneumatic/Hydraulic                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Pre-identification | When preparing the work, its aim is to ensure that the work will be carried out on the structure or installation to be locked out.                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Separation         | Separate from all sources of tension. It must be done with certainty. Separation must be carried out by means of devices provided for this purpose on all active conductors, including the distributed neutral, with the exception of the protective and neutral conductor (PNC). Switching off all power and control circuits is done in a fully visible manner, including emergency power supplies. | Separate the work equipment from any source of energy using devices provided for this purpose, by uncoupling each coupling system (clutches, etc.). Cutting off the transmission of all forms of energy in a fully visible manner, including emergency and energy accumulators.                               | Separate all upstream and downstream fluid sources by:<br><b>Simple isolation</b> (locking a single device or switching element into the closed position)<br><b>Reinforced isolation</b> on all possible fluid inlets (by interposition of a full seal, by clear interruption of the piping, or by maintaining in the closed position of two successive cut-off devices mounted in series with the open position of the purge).<br><br>Electrical instructions are required beforehand. |
| Condemnation       | Prevent the operation of the separation device, obtained by locking using a material device that is difficult to neutralize so that its removal is impossible without the voluntary action of an authorized person.                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                               | Lock the separation devices in the closed position and the purge devices in the open position. Equip the equipment with separation devices equipped with integrated locking means (padlockable valves or purges, etc.).                                                                                                                                                                                                                                                                 |
|                    | <b>Named padlock</b> – the person who locks out the equipment/system affixes a named security padlock with a unique key to each energy lockout point.<br><b>Anyone</b> working on a locked system applies his own security padlock protecting it from untimely restarting either on the blocked unit or on a lockout box.                                                                             |                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Dissipation        |                                                                                                                                                                                                                                                                                                                                                                                                       | Evacuate any residual mechanical energy in the work equipment, for example by relaxing a spring or a metal cable, or wait for a motor to completely stop.<br>If it is impossible to ensure complete dissipation of the residual energy, immobilise the mechanical elements using suitable clamping equipment. | Dissipate by physical elimination of potential or residual energies or to evacuate fluids, by purging device lockable in the immobilised position and lockable with a key<br><br>In the event that elimination is not possible, it is possible to resort to retention or confinement of energies.                                                                                                                                                                                       |
| Signaling          | Clear and permanent information on the completion of the condemnation. A label or signs are affixed indicating that the installation has been de-energized after verifying that this was indeed the case. Report that the device/part cannot be operated.                                                                                                                                             |                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Identification     | Allows to be certain that the work is carried out on the structure or installation separated from any power supply by separation devices locked in the open position. This is the verification of the key points of the electrical circuits                                                                                                                                                           |                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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|                        | Electric                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Mechanical                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                             |
|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Machine stopped / moving                                                                                                                                                                                                                                                                   | Pneumatic/Hydraulic                                                                                                                                                                                                                                                                                                         |
| Verification           | Check the absence of voltage before any intervention on an installation or electrical equipment. It is prohibited to use a measuring device to carry out the electrical no Voltage Check. Use of a standardized device specially designed for this purpose. Carry out your self-check before and after use.                                                                                                                                                                                                                                                           | Check the absence of risks linked to the movement of the mechanisms by visual inspection of dissipation and immobilization. Absence of energy (tension, pressure, movement)                                                                                                                | Check the absence of residual risk, the effective absence of energy or fluid. The absence of flow is only an indication because a product may be congealing, clogging, crystallizing or viscous. This is why the effectiveness of the purge must be checked and the locking valves checked to confirm them.-as non-elusive. |
| Grounding and Shorting | If required in LV, earth the electrical installation immediately after checking the absence of voltage by causing a short circuit of all active conductors including the neutral when it is distributed. It protects against possible voltage re-supplies from upstream or downstream or from independent sources. It also helps protect against the effects of magnetic induction and capacitive coupling. It causes a cut-off by short circuit with dissipation by earth<br>The grounding is chosen according to the short-circuit current at the attachment point. |                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                             |
| Reinstatement          | Reinstatements are generally carried out in the reverse order of lockout operations. Remove any screens, protectors, markings, etc.; remove the short circuits then the earth connections; open the grounding disconnectors or switches if they are used; withdraw the conviction of the separation body; close the separation member.                                                                                                                                                                                                                                | The risk analysis determines the order and content of operations for a safe restart. Remove the mechanical immobilizing elements (pins, jack stands, or other clamping devices) by implementing compensatory measures such as procedure, means of communication, specific training, etc.). | Ensure the real end of operations (assurance of proper reassembly, safety of those involved). The risk analysis determines the order and content of operations for restarting in complete safety, such as removing or stopping the dissipation device, by initializing the equipment controlled by automation.              |